

**STATISTIC COMPUTATION
COMPOUND LINIER REGRESSION**



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STEP 1 – DATA COLLECTION

The data that used for this answers is car price data prediction from Kaggle on this link:

<https://www.kaggle.com/code/muhammadglennyunifer/car-price-prediction-linear-regression/notebook>

And here the link for spreadsheet results:

https://docs.google.com/spreadsheets/d/1ihEAN6-tDojuW7mEO5Zo8E37xuTbdW_uBS2JKoOgvPM/edit?usp=sharing

STEP 2 – VARIABLE SELECTION

- **Dependent Variable**

- Price

- **Independent Variable**

- | | |
|-------------------|--------------------|
| ▪ Symboling | ▪ EngineeType |
| ▪ Fueltype | ▪ CylinderNumber |
| ▪ Aspiration | ▪ EngineeSize |
| ▪ DoorNumber | ▪ FuelSystem |
| ▪ CarBody | ▪ BoreRatio |
| ▪ DriveWheel | ▪ Stroke |
| ▪ EngineeLocation | ▪ CompressionRatio |
| ▪ WheelBase | ▪ HorsePower |
| ▪ CarLength | ▪ PeakRPM |
| ▪ CarWidth | ▪ CityMpg |
| ▪ CarHeight | ▪ HighwayMpg |
| ▪ CurbWeight | |

STEP 3 – REGRESSION ANALYSIS & INTERPRETATION RESULTS

In this variable relationship analysis. I analyzed the variable relationships that affect the value of the CityMPG variable. The following are the variables that are thought to affect the value of the CityMPG variable

- **EngineType**

EngineType & CityMPG								
Regression Statistics								
Multiple R	0,1786041491							
R Square	0,03189944209							
Adjusted R Squ	0,02710686507							
Standard Error	6,462107067							
Observations	204							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	277,9471883	277,9471883	6,656010317	0,01059176511			
Residual	202	8435,283204	41,75882774					
Total	203	8713,230392						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	27,66294592	1,04238538	26,53811772	0	25,60759405	29,71829779	25,60759405	29,71829779
0	-1,033976921	0,40077798	-2,579924479	0,01059176551	-1,82422186	-0,2437319827	-1,82422186	-0,2437319827

The obtained regression results must be analyzed by observing the slope, intercept, coefficient of determination (R^2), and the evaluation metric (MSE). The **intercept** value of 27.66 suggests that when the EngineType variable is at its baseline category (e.g., zero), the predicted CityMPG is approximately 27.66. The **slope coefficient** of -1.03 indicates that a change in EngineType is associated with a decrease in CityMPG by around 1.03 miles per gallon, suggesting a negative relationship. The **R Square (R^2)** value of 0.0319 implies that only about 3.2% of the variation in CityMPG can be explained by the EngineType variable, meaning the model has low explanatory power. The **Mean Squared Error (MSE)**, calculated as the residual Mean Square (MS) value of 41.76, shows the average squared difference between the actual and predicted CityMPG values — a relatively high value, indicating limited prediction accuracy. Despite the low R^2 , the **p-value** of 0.0106 and **Significance F** under 0.05 indicate that EngineType has a statistically significant effect on CityMPG.

- **CylinderNumber**

CylinderNumber & CityMPG								
Regression Statistics								
Multiple R	0,4475508832							
R Square	0,2003017931							
Adjusted R Squ	0,196342891							
Standard Error	5,873229551							
Observations	204							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	1745,275671	1745,275671	50,59528938	0			
Residual	202	6967,954721	34,49482535					
Total	203	8713,230392						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	37,10307072	1,71771154	21,60029194	0	33,71612603	40,4900154	33,71612603	40,4900154
4	-2,706964683	0,3805638636	-7,113036579	0	-3,457351885	-1,956577481	-3,457351885	-1,956577481

The regression results between the variables **cylindernumber** and **citympg** show a significant relationship with an **R²** value of 0.2003, meaning 20.03% of the variation in **citympg** can be explained by the **cylindernumber** variable. The **intercept** value of 37.103 indicates that when **cylindernumber** is 0, the estimated **citympg** is 37.103. Meanwhile, the **slope** (coefficient for **cylindernumber**) of -2.707 shows that for each increase of one unit in **cylindernumber**, **citympg** is expected to decrease by 2.707. The **p-value** for both the intercept and slope is less than 0.05, indicating that the relationship between these two variables is statistically significant. The low **R²** value suggests that this model only explains a small portion of the variation in **citympg**, indicating that other variables may have a stronger impact on **citympg**. With a high **F-statistic** (50.595) and a very low **p-value**, the overall regression model is considered significant. However, the relatively large standard error (5.873) indicates that predictions from this model may have considerable deviation.

- **FuelSystem**

FuelSystem & CityMPG								
Regression Statistics								
Multiple R	0,2599914999							
R Square	0,06759558003							
Adjusted R Squ	0,06297971657							
Standard Error	6,341851609							
Observations	204							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	588,9758623	588,9758623	14,64418966	0,000173071516			
Residual	202	8124,25453	40,21908183					
Total	203	8713,230392						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	23,30454626	0,6730557343	34,62498731	0	21,97743021	24,6316623	21,97743021	24,6316623
1	0,778841348	0,203524327	3,826772747	0,000173071516	0,3775366922	1,180146004	0,3775366922	1,180146004

The regression results between the variables **fuelsystem** and **citympg** show a modest relationship, with an **R²** value of 0.0676, indicating that only 6.76% of the variation in **citympg** can be explained by **fuelsystem**. The **intercept** value of 23.305 suggests that when **fuelsystem** is 0, the estimated **citympg** is 23.305. The **slope** (coefficient for **fuelsystem**) of 0.779 indicates that for each increase of one unit in **fuelsystem**, **citympg** is expected to increase by 0.779. The **p-value** for both the intercept and the slope is less than 0.05, which means that the relationship between the two variables is statistically significant. However, the low **R²** value suggests that this model only explains a small portion of the variation in **citympg**, indicating that other factors may have a greater influence on **citympg**. Despite the relatively small proportion of variation explained by this model, the high **F-statistic** (14.644) and low **p-value** (0.00017) suggest that the model is statistically significant. The standard error of 6.342 indicates that predictions from this model may have considerable variability.

- **HorsePower**

HorsePower & CityMPG									
Regression Statistics									
Multiple R	0,8017848608								
R Square	0,642858963								
Adjusted R Squ	0,6410909381								
Standard Error	3,924946955								
Observations	204								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	5601,378255	5601,378255	363,6028826	0				
Residual	202	3111,852138	15,4052086						
Total	203	8713,230392							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%	
Intercept	39,0333486	0,7737921617	50,44422847	0	37,50760275	40,55909444	37,50760275	40,55909444	
111	-0,1325202804	0,00694974208	-19,06837388	0	-0,1462236243	-0,1188169364	-0,1462236243	-0,1188169364	

The regression results between the variables **horsepower** and **citympg** show a strong relationship with an **R²** value of 0.6429, meaning that 64.29% of the variation in **citympg** can be explained by **horsepower**. The **intercept** value of 39.033 indicates that when **horsepower** is 0, the estimated **citympg** is 39.033. The **slope** (coefficient for **horsepower**) of -0.133 shows that for each increase of one unit in **horsepower**, **citympg** is expected to decrease by 0.133. The **p-value** for both the intercept and slope is less than 0.05, indicating that the relationship between **horsepower** and **citympg** is statistically significant. The relatively high **R²** value suggests that the model explains a significant portion of the variation in **citympg**, indicating that **horsepower** is a substantial factor in determining **citympg**. The high **F-statistic** of 363.603 and the very low **p-value** (0) also suggest that the overall regression model is statistically significant. The standard error of 3.925 indicates that predictions from this model have relatively low variability compared to the magnitude of the estimates.